

FILE2EDI

Recommendations for Optimizing the SHIPTO Validation Flow

Purchase order PDF -> customer identification -> SHIPTO resolution -> address validation -> EDIFACT

Document objective

Transform the current flow, which is very complete but still too decision-tree oriented, into an industrializable logic: candidate generation, explainable scoring, controlled LLM fallback, anti-false-positive post-validation and traceable final decision.

1. Executive summary

The current schema is functionally strong: it avoids validating a SHIPTO only because it belongs to the SOLDTO, requires documentary evidence and includes human validation when there is doubt.

The recommended optimization is to replace the decision tree with a scoring engine. The workflow becomes easier to maintain, less dependent on client PDF layouts and easier to audit in production.

Target principle

Deterministic rules first, LLM only as a fallback on the 2 or 3 best candidates, followed by blocking post-validation before EDIFACT generation.

2. Assessment of the current schema

Strengths

- Clear separation between SOLDTO customer identification and SHIPTO delivery-location selection.
- Address post-validation is included to reduce false positives.
- Ambiguous cases are explicitly handled: multi-SHIPTO, incomplete address, missing postal code and uncertain LLM output.
- Strong safety logic: when there is doubt, the order is not automatically sent as EDIFACT.

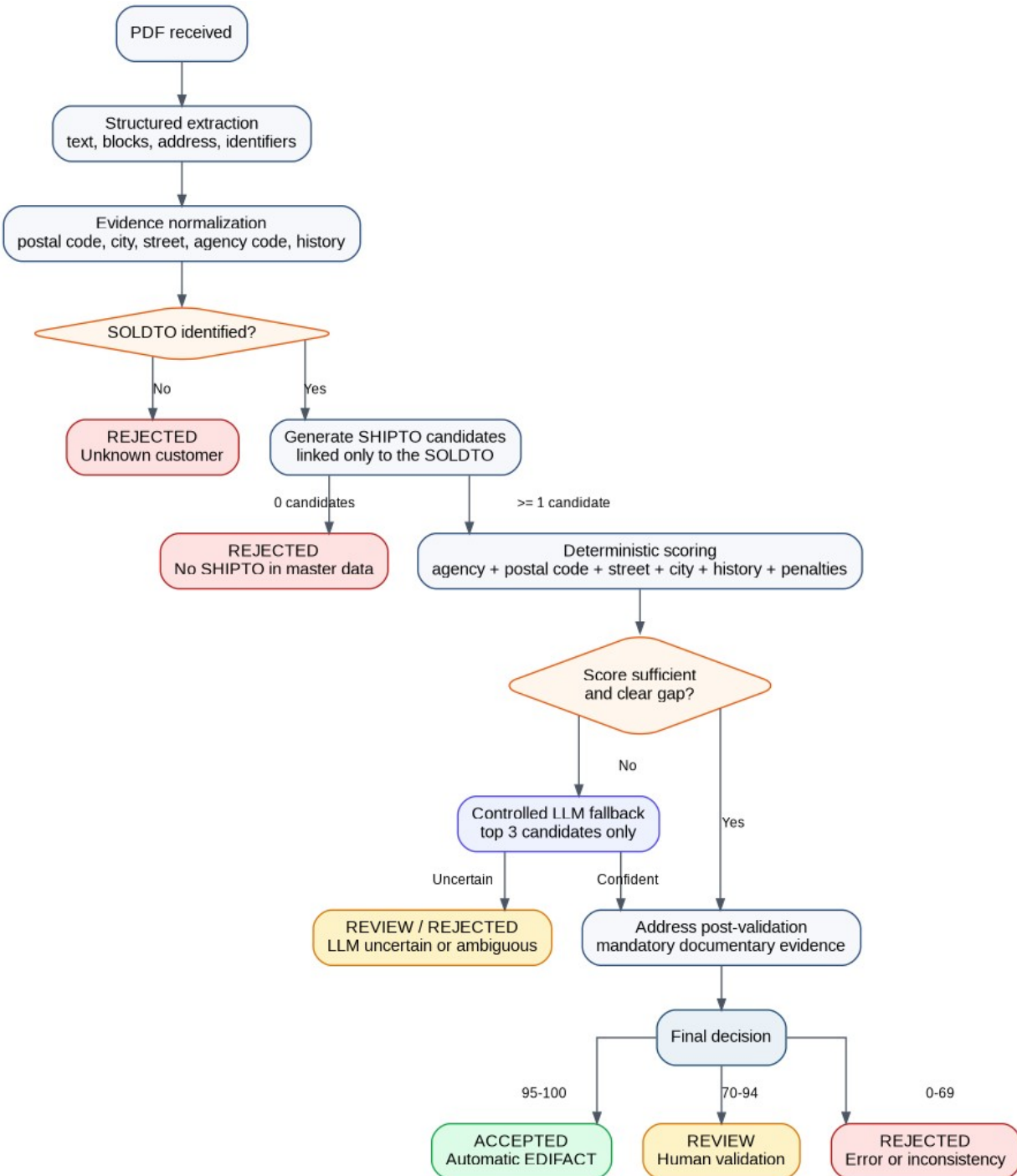
Limitations to fix

- Too many conditional branches: the schema may become hard to maintain with many customers and PDF formats.
- The LLM is involved in several places, which can increase cost, latency and variability.
- The same score may represent different situations; for example, 80% can mean a unique SHIPTO or a partially confirmed address.
- The system lacks normalized reason codes to automatically explain each decision.

3. Optimized target schema

The target schema replaces successive decision levels with a six-step approach:

1. Extract and normalize all evidence from the PDF.
2. Identify the SOLDTO with a dedicated confidence score.
3. Generate all SHIPTO candidates linked to the SOLDTO.
4. Score each candidate using the detected evidence.
5. Use the LLM only when the best score is not sufficient or when the gap with the second candidate is too small.
6. Apply blocking post-validation before automatic EDIFACT generation.



4. SHIPTO scoring engine

Main recommendation: each SHIPTO candidate receives a score based on evidence found in the PDF. Strong evidence adds points, while contradictions add penalties.

Evidence / signal	Score	Comment
SHIPTO postal code present in the PDF	+40	Strong evidence, easy to trace
Normalized street present in the PDF	+30	Strong evidence if OCR is reliable
SHIPTO city present in the PDF	+20	Medium evidence, useful as a complement
Agency code detected and linked to the SHIPTO	+35	Very useful for customers with CAC / agency references
Order or history already associated with the SHIPTO	+20	Logical evidence, not sufficient alone
Agency name or customer name is close	+10	Weak signal
Postal code of another SHIPTO is better represented	-40	Contradiction signal
Clearly contradictory address	-50	Likely blocking issue
No address evidence at all	-30	Prevents automatic acceptance

Proposed selection rule

A candidate can be automatically accepted only if its score is ≥ 95 , the gap with the second candidate is sufficient, and address post-validation finds no inconsistency.

5. Separating SOLDTO and SHIPTO confidence

Two separate confidence values should be stored:

- soldto_confidence: quality of the customer/order-giver identification.
- shipto_confidence: quality of the delivery-address resolution.

Expected output example:

```
{
  "soldto": {
    "id": "123456",
    "confidence": 98,
    "matched_by": ["VAT", "SIREN"]
  },
  "shipto": {
    "id": "987654",
    "confidence": 86,
    "matched_by": ["postal_code", "city", "llm_validation"]
  },
  "final_decision": "REVIEW",
  "reason_codes": [
    "SOLDTO_FOUND_BY_VAT",
    "SHIPTO_CONFIRMED_BY_CITY",
    "POSTAL_CODE_NOT_FOUND_DIRECTLY"
  ]
}
```

6. Recommended decision grid

Score	Decision	Action	Interpretation
95-100	ACCEPTED	Automatic EDIFACT generation	Strong evidence + post-validation OK
85-94	Light REVIEW	Fast validation	Good candidate but indirect or incomplete evidence
70-84	Mandatory REVIEW	Human control	Plausible but ambiguous candidate
40-69	Functional REJECTED	Mandatory human intervention	Insufficient data
0-39	Critical REJECTED	Error or inconsistency	Contradictory customer/address

7. Controlled use of the LLM

The LLM must not replace business rules. It should only be used when deterministic evidence is insufficient.

- Never ask the LLM to choose from the full SHIPTO master data.
- Limit LLM arbitration to the top 2 or 3 candidates scored by the rules.
- Provide only useful evidence to the LLM: PDF excerpt, candidate address, score, positive reasons and contradictions.
- Require a structured output: decision, confidence, short justification, confirmed fields and missing fields.
- If the LLM is uncertain, route to REVIEW or REJECTED, never ACCEPTED.

Element	Rule
LLM input	Top 3 SHIPTO candidates + evidence detected in the PDF + contradictions
LLM output	decision, confidence, selected_shipto_id, justification, missing_evidence
Forbidden	No automatic selection if the LLM returns uncertain / insufficient evidence
Purpose	Disambiguate close candidates, not invent an address or customer

8. Anti-false-positive post-validation

Post-validation remains mandatory after selecting the best SHIPTO. It acts as the final guardrail before EDIFACT generation.

- If postal code + street or city are confirmed in the PDF: acceptance may be possible.
- If only an indirect signal exists: REVIEW.
- If no address element confirms the SHIPTO: REJECTED or blocking REVIEW.
- If a contradictory address is detected: REJECTED, even if the SOLDTO is reliable.

Fundamental rule

A SHIPTO can never be accepted only because it belongs to the SOLDTO. It must be confirmed by at least one delivery evidence item: agency code, postal code, street, city, order history or controlled LLM validation.

9. Master data and index optimization

To make the system fast and maintainable, dedicated indexes should be prepared before execution.

Index	Purpose
partners_by_soldto	List all SHIPTO records linked to a SOLDTO
partners_by_postal_code	Find candidates quickly by postal code
partners_by_agency_code	Use CAC / agency-code references
partners_by_normalized_city	Compare cities without accents or spelling variants
partners_by_normalized_street	Compare normalized streets
order_history_by_customer	Strengthen scoring using customer/order history

Recommended normalization:

- Remove accents, enforce consistent casing and clean punctuation.
- Normalize abbreviations: St -> Saint, Av -> Avenue, Blvd -> Boulevard.
- Tokenize addresses to compare street, city and postal code separately.
- Keep both raw and normalized values for auditability.

10. n8n / Databricks integration recommendations

Given your `/api/extract/base64` endpoint, the n8n integration can be simplified:

- Remove the Extract PDF text node if the Databricks API extracts the PDF itself.
- Keep the Normalize + Hash PDF node: it checks the %PDF- magic bytes, generates the SHA-256 hash, sanitizes the filename and produces pdf_base64.
- First call `/oidc/v1/token` to obtain an OAuth access_token, then call `/api/extract/base64` in JSON.
- Send filename + content_base64, and log correlation_id, execution_id, pdf_hash and response status.
- Never store a Databricks PAT directly in the workflow; use n8n credentials or environment variables.

11. Implementation roadmap

Step	Action	Deliverable
1	Stabilize the API output	Single JSON response: status, decision, scores, reason_codes, edifact, errors
2	Build master-data indexes	partners_by_soldto, agency, postal code, city, normalized street
3	Add candidate scoring	Compute score + reason_codes for each SHIPTO
4	Control the LLM fallback	Top 3 candidates only + strict output schema
5	Add blocking post-validation	Block if there is no address evidence or if contradictions exist
6	Connect n8n production flow	OAuth token + <code>/api/extract/base64</code> + logs + error handling
7	Measure quality	Accepted/review/rejected rate, false positives, average time, LLM cost

12. KPIs and logging

- Automatic ACCEPTED rate.
- REVIEW rate by reason: missing address, multi-SHIPTO, uncertain LLM, unknown customer.
- REJECTED rate by critical reason.
- Average processing time per PDF.
- Percentage of orders using the LLM.
- Average gap between the best and second-best SHIPTO candidate.

- Number of false positives detected after human validation.

13. Final recommendation

Recommended position

Keep the current schema as business documentation, but implement the target engine as candidate generation + scoring + LLM fallback + blocking post-validation. This is the most robust approach to industrialize FILE2EDI across many customers and PDF formats.

End of document.